

Caesar Cipher

1. Write a C program for Caesar cipher involves replacing each letter of the alphabet with the letter standing k places further down the alphabet, for k in the range 1 through 25.

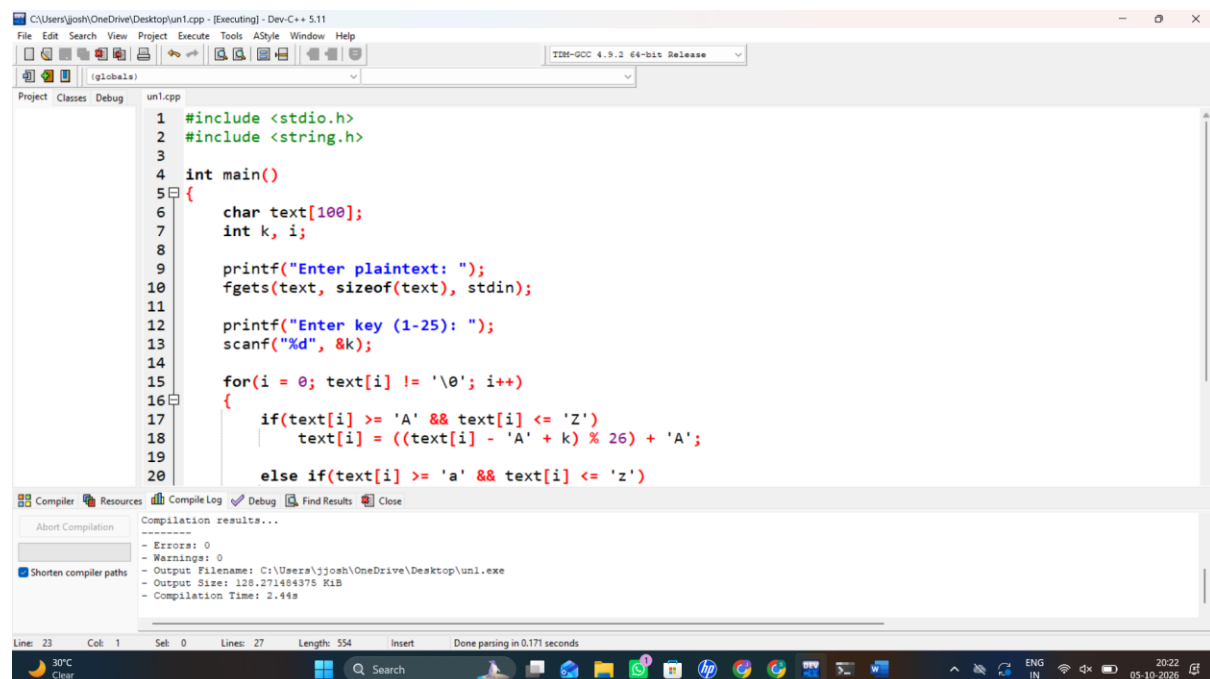
Aim

To write a C program to implement the **Caesar Cipher**, in which each letter of the alphabet is replaced by the letter located k positions further in the alphabet.

Algorithm

1. Start the program.
2. Read the plaintext and the key k.
3. For each character in the plaintext:
 - If it is an uppercase letter, shift it by k positions.
 - If it is a lowercase letter, shift it by k positions.
 - Preserve spaces and other characters.
4. Display the encrypted ciphertext.
5. Stop.

Program



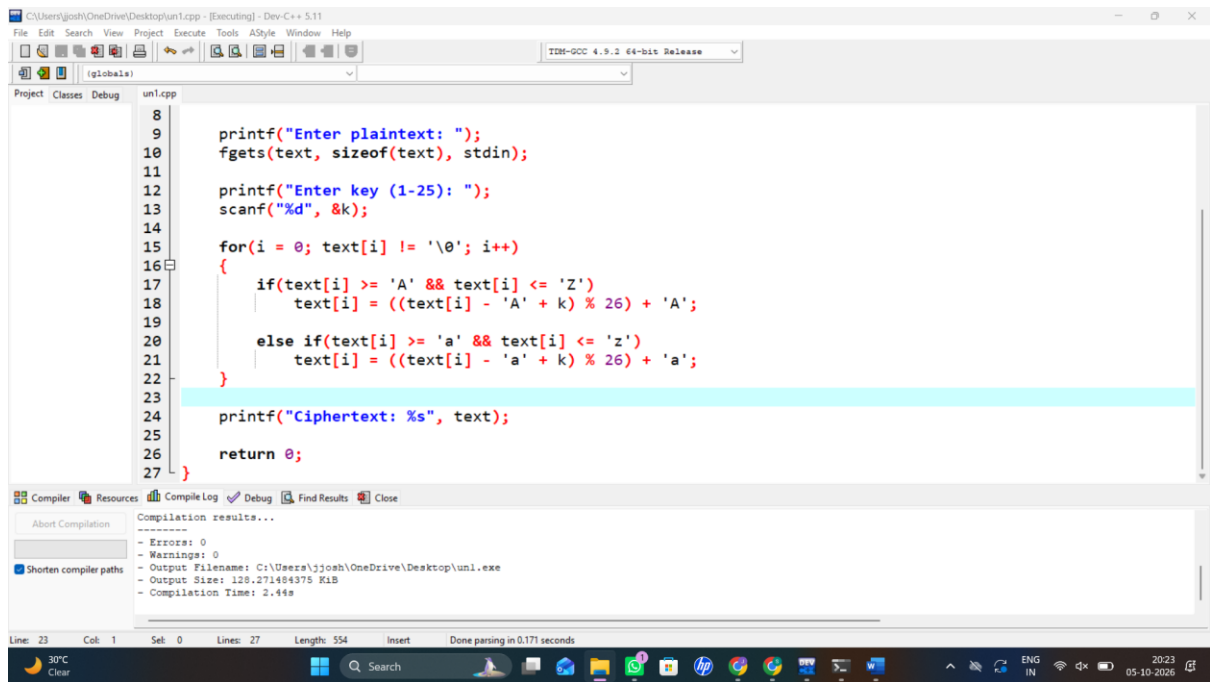
The screenshot shows a Dev-C++ IDE window with a C program for the Caesar Cipher. The code is as follows:

```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main()
5 {
6     char text[100];
7     int k, i;
8
9     printf("Enter plaintext: ");
10    fgets(text, sizeof(text), stdin);
11
12    printf("Enter key (1-25): ");
13    scanf("%d", &k);
14
15    for(i = 0; text[i] != '\0'; i++)
16    {
17        if(text[i] >= 'A' && text[i] <= 'Z')
18            text[i] = ((text[i] - 'A' + k) % 26) + 'A';
19
20        else if(text[i] >= 'a' && text[i] <= 'z')
```

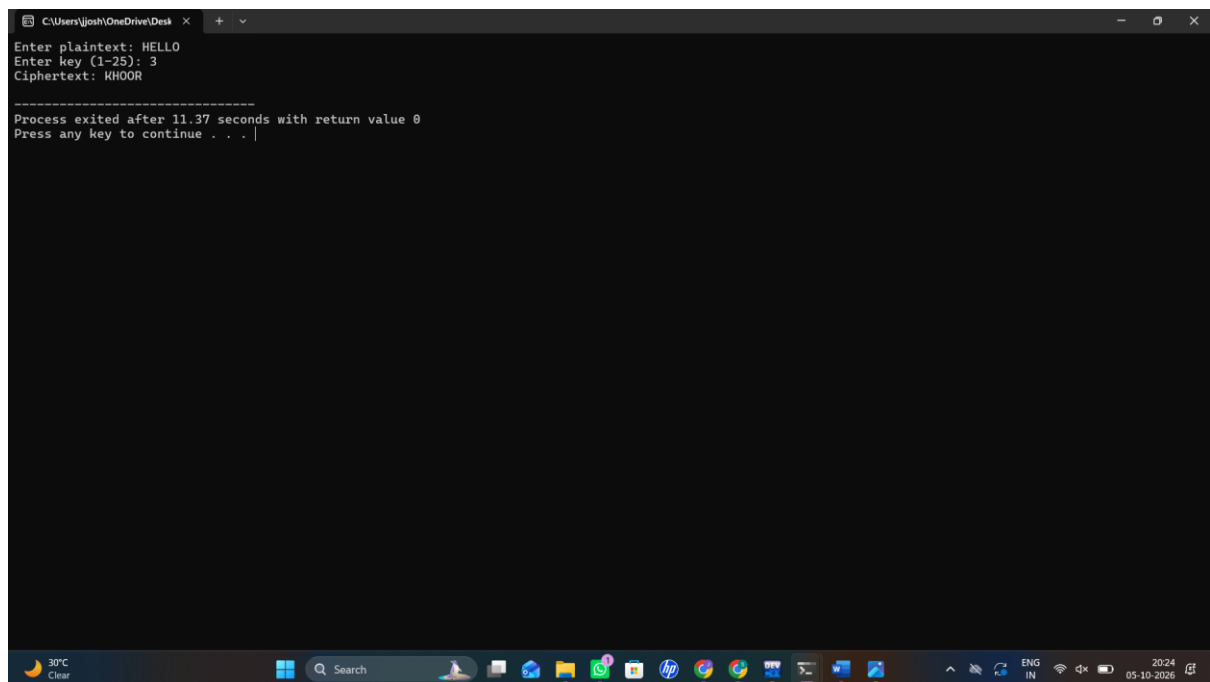
The bottom panel shows the compilation results:

```
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\josh\OneDrive\Desktop\unl.exe
- Output Size: 128.271484375 KiB
- Compilation Time: 2.44s
```

The status bar at the bottom indicates the file is 'unl.cpp', line 23, column 1, with 27 lines and 554 characters. The system tray shows the date as 05-10-2026 and time as 20:22.



Input/Output



Result

Thus, the **Caesar Cipher program** was successfully implemented using a key k .